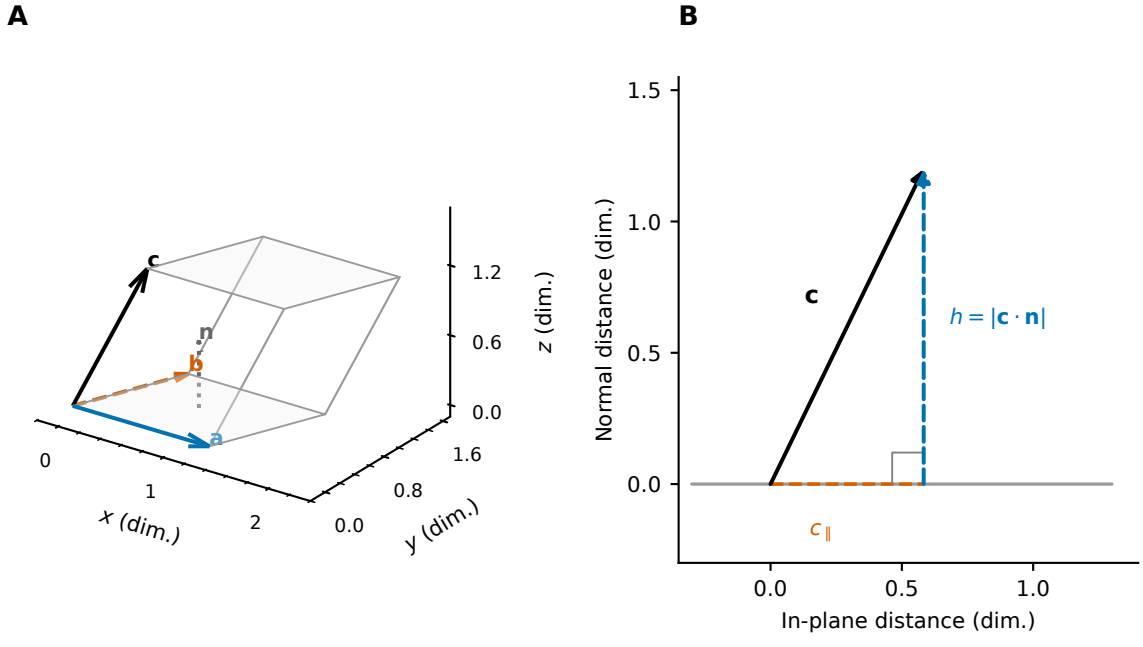

$A = \|\mathbf{a} \times \mathbf{b}\|$; $V = |(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}|$

Only normal height contributes: $V = A\,h$